

NUMBER SEQUENCES AND SERIES

Lesson 26

Defn (Sequence)

A sequence is an ordered pattern of numbers, defined by any function $f(n)$, whose inputs are strictly positive or negative integers ($n \in \mathbb{Z}$) and output in the form of real values ($\mathbb{R}$).

Notation: $a_n = \{f(n)\}_{n=1}^{\infty}$, where $f: \mathbb{Z} \rightarrow \mathbb{R}$
 $\leftarrow$ ending input
 $\leftarrow$ starting input

Examples of Sequences

(i) $a_n = \{3-2n\}_{n=1}^{\infty} = 1, -1, -3, -5, -7, \dots$ (an A.P sequence)
 $\uparrow f(1) \quad \uparrow f(2) \quad \uparrow f(3) \quad \uparrow f(4) \quad \uparrow f(5)$

(ii) $b_n = \{4 \cdot (\frac{1}{2})^{n-1}\}_{n=1}^{\infty} = 4, 2, 1, \frac{1}{2}, \dots$ (a G.P sequence)

(iii) $c_n = \{\frac{1}{n}\}_{n=1}^{\infty} = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$ (Harmonic sequence)

(iv) $d_n = \{\frac{1}{n(n+1)}\}_{n=1}^{\infty} = \frac{1}{2}, \frac{1}{2 \cdot 3}, \frac{1}{3 \cdot 4}, \frac{1}{4 \cdot 5}, \dots$ (Telescopic sequence)

(v) $e_n = \{\frac{n(n+1)}{2}\}_{n=1}^{\infty} = 1, 3, 6, 10, 15, \dots$ (Sequence of triangular no's)

(vi) $f_n = \{n^2\}_{n=1}^{\infty} = 1, 4, 9, 16, 25, 36, \dots$ (sequence of Square no's).

Defn (Series)

A series is a sum of all terms of a sequence.

Notation

$$\sum_{n=1}^{\infty} a_n$$

Examples of Series

(i) $\sum_{n=1}^{\infty} 4(\frac{1}{2})^{n-1} = 4 + 2 + 1 + \frac{1}{2} + \dots$ (Series of b_n -sequence)

(ii) $\sum_{n=1}^{\infty} \frac{n(n+1)}{2} = 1 + 3 + 6 + 10 + 15 + \dots$ (Series of d_n -sequence).

ARITHMETIC SEQUENCE & SERIES

Defn: An Arithmetic progression (A.P) sequence with starting term 'a' and stepsize 'd' (common difference) is given by:

AP-Sequence: $T_n = a + d(n-1) \leftarrow n^{\text{th}} \text{ term of A.P}$

Arithmetic ~~Sequence~~

Sequence: $a, a+d, a+2d, a+3d, a+4d, \dots, a+d(n-1), \dots$
 $\xrightarrow{T_1} \xrightarrow{T_2} \xrightarrow{T_3} \xrightarrow{T_4} \xrightarrow{T_5} \xrightarrow{T_n}$
 first term $+d$ $+d$ $+d$ $+d$ $\leftarrow$ (Step Size adds d)

AP SERIES: Is the sum of all terms of AP up to the n^{th} term

$$S_n = \underbrace{a}_{T_1} + \underbrace{a+d}_{T_2} + \underbrace{a+2d}_{T_3} + \underbrace{a+3d}_{T_4} + \underbrace{a+4d}_{T_5} + \dots + \underbrace{a+d(n-1)}_{T_n}$$

$$S_n = \frac{n}{2} [2a + d(n-1)] \leftarrow \text{Series of A.P up to } n^{\text{th}} \text{ term}$$

Example 1

The 7th term of an AP sequence is 22 and the 12th term is 37. Find the sum of the first 20 terms.

Solution

n^{th} term of A.P: $T_n = a + d(n-1)$

7th term: $T_7 = a + 6d = 22$

12th term: $T_{12} = a + 11d = 37$

Solving $\begin{cases} a + 6d = 22 \text{ --- (1)} \\ a + 11d = 37 \text{ --- (2)} \end{cases}$

$\therefore 5d = 15, \therefore d = 3$
 $\therefore a = 4$

Sum of AP: $S_n = \frac{n}{2} [2a + d(n-1)]$

Sum of 1st

20 terms: $S_{20} = \frac{20}{2} [2(4) + 3(19)] = 650$

$n = 20$

$a = 4$

$d = 3$

$\therefore S_{20} = 650 \leftarrow \text{Sum of 1st 20 terms.}$

③

GEOMETRIC PROGRESSION (G.P) - SEQUENCE & SERIES

Defn: A Geometric progression (G.P) sequence with starting term "a" and common ratio "r" is given by:

G.P sequence: $T_n = ar^{n-1}$ $\leftarrow$ n^{th} term of G.P.

G.P sequence: $a, ar, ar^2, ar^3, ar^4, \dots, ar^{n-1}, \dots$

$\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & & \uparrow \\ \text{1st term} & T_1 & T_2 & T_3 & T_4 & T_5 & T_n \\ & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\ & \times r & \times r & \times r & \times r & & \end{matrix}$

G.P SERIES: Is the sum of all terms of G.P up to the n^{th} term.

$$S_n = \underbrace{a}_{T_1} + \underbrace{ar}_{T_2} + \underbrace{ar^2}_{T_3} + \underbrace{ar^3}_{T_4} + \underbrace{ar^4}_{T_5} + \dots + \underbrace{ar^{n-1}}_{T_n}$$

$$\underline{S_n = \frac{a(1-r^n)}{1-r}} \quad \leftarrow \text{Series of G.P up to } n^{\text{th}} \text{ term}$$

Example 1

The 5th term of a G.P sequence is 162 and the 8th term is 4374. Find the sum of the first 10 terms.

Solution:

n^{th} term of G.P. is $T_n = ar^{n-1}$

5th term: $T_5 = ar^4 = 162$ — (1) } Solving $\frac{ar^7}{ar^4} = \frac{4374}{162} = 27$

8th term: $T_8 = ar^7 = 4374$ — (2) } $\frac{(2)}{(1)} \quad r^3 = 27, \therefore r = 3$

$\therefore a = 2$

Sum of G.P: $S_n = \frac{a(1-r^n)}{1-r}$

Sum of first 10 terms: ie $S_{10} = \frac{2(1-3^{10})}{1-3} = 59,048$

$n=10$

SERIES OF POWERS OF THE NATURAL NUMBERS

(4)

Recall natural numbers ($\mathbb{N}$) i.e. 1, 2, 3, 4, 5, 6, 7, ...

Sequence $\{r\}_{r=1}^n = 1, 2, 3, 4, 5, \dots, n$ (Sequence of n , natural no's)

SERIES:

$$(1) \sum_{r=1}^n r = 1 + 2 + 3 + 4 + 5 + \dots + n = \frac{n(n+1)}{2}$$

$$\boxed{\sum_{r=1}^n r = \frac{n(n+1)}{2}}$$

Sum of natural no's.

$$(2) \sum_{r=1}^n r^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\boxed{\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}}$$

Sum of Squares
of natural no's.

$$(3) \sum_{r=1}^n r^3 = 1^3 + 2^3 + 3^3 + 4^3 + 5^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$\boxed{\sum_{r=1}^n r^3 = \frac{n^2(n+1)^2}{4}}$$

Sum of Cubes
of natural no's

Example 1

Find the sum of the series $\sum_{r=1}^5 r(3+2r)$.

Solution

$$\sum_{r=1}^5 r(3+2r) = 3 \sum_{r=1}^5 r + 2 \sum_{r=1}^5 r^2$$

$$= 3 \left[\frac{n(n+1)}{2} \right] + 2 \left[\frac{n(n+1)(2n+1)}{6} \right]$$

$$\boxed{n=5}$$

$$= 3 \left(\frac{5(6)}{2} \right) + 2 \left[\frac{5(6)(11)}{6} \right] = \underline{\underline{155}}$$

Example 2

-5-

Evaluate the following series

$$(a) S = 1^2 + 3^2 + 5^2 + \dots + 601^2$$

$$(b) T = 1^3 + 6^3 + 11^3 + \dots + 236^3$$

Solution

$$a=1, d=2$$

$$(a) AP: \underbrace{1, 3, 5, \dots}_{+2 \quad +2}, 601 \text{ ie } T_n = 1 + 2(n-1) = 2n-1$$

$$\text{Last Term: } 2N-1=601$$

$$2N=602 \quad \therefore N = \underline{301}$$

$$S = \sum_{r=1}^{301} (2r-1)^2 = \sum_{r=1}^{301} (4r^2 - 4r + 1)$$

$$= 4 \sum_{r=1}^{301} r^2 - 4 \sum_{r=1}^{301} r + \sum_{r=1}^{301} 1$$

$$= 4 \frac{(301)(302)(603)}{6} - 4 \frac{(301)(302)}{2} + 1 \cdot (301)$$

$$= 36,542,604 - 181,804 + 301 = \underline{\underline{36,361,101}}$$

$$(b) AP: \underbrace{1, 6, 11, \dots}_{+5 \quad +5}, 236^3 \text{ ie } T_n = 1 + 5(n-1) = 5n-4$$

$$a=1, d=5$$

$$\text{Last term: } 5N-4=236$$

$$5N=240 \quad \therefore N = \underline{48}$$

$$T = \sum_{r=1}^{48} (5r-4)^3 = \sum_{r=1}^{48} (125r^3 - 300r^2 + 240r - 64)$$

$$= 125 \sum_{r=1}^{48} r^3 - 300 \sum_{r=1}^{48} r^2 + 240 \sum_{r=1}^{48} r - \sum_{r=1}^{48} 64$$

$$= 125 \frac{(48)^2(49)^2}{4} - 300 \frac{(48)(49)(97)}{6} + 240 \frac{(48)(49)}{2} - 64(48)$$

$$= 172,872,000 - 11,407,200 + 282,240 - 3,072$$

$$= \underline{\underline{161,743,968}}$$